ScholarMind-AI

Intelligent Research Paper Analysis, Knowledge Discovery and Academic Recommendation Platform

Major Project – Stage-I Review-1 Presentation

by

Name: G Amulya **S Ashritha**
Roll No: 23B81A05D3 **23B81A05E0**

Section:CSE-C
Batch No:
Dept:Computer Science and Engineering
CVR College of Engineering

Under the Esteemed guidance of

Supervisor name: Ms B.Sharmila
Designation: Asst. Professor

Abstract

- Students, researchers, and academicians often face difficulties in discovering reliable academic resources, understanding complex research content, and identifying research opportunities. **ScholarMind AI** aims to provide an intelligent platform for research assistance, knowledge discovery, academic recommendations, and personalized learning support.
- The system allows users to search academic resources, upload and analyze research papers, generate summaries, extract key insights, compare documents, and interact with research content through AI-based question answering. It also supports semantic search, related-topic discovery, research-gap identification, and personalized recommendations..
- By combining AI, NLP, LLMs, semantic search, and recommendation techniques, ScholarMind AI provides a centralized academic research environment that reduces the time required to search and analyze information while helping users make better academic and research decisions.

Agenda

1. Aim, motivation, novelty
2. problem statement, objectives, and scope
3. Literature review
4. Research gaps and proposed solution.
5. Software requirements (Software and Hardware requirements)
6. Plan project milestones.
7. Design and overall system architecture.
   - Architecture
   - modules

1.Aim, motivation, novelty

- **Aim**

To develop ScholarMind AI, an intelligent research platform that uses AI, LLMs, and RAG to help students and researchers discover, understand, analyze, and organize research papers, including AI-powered summaries, key findings, paper analysis, and research assistance.

- **Motivation**

Students and researchers often spend significant time searching, reading, and understanding lengthy research papers using multiple tools. ScholarMind AI aims to bring these activities into one intelligent platform, making research faster, easier, and more organized through AI-powered analysis and document-based assistance.

- **Novelty**

The novelty of ScholarMind AI lies in combining AI-powered research-paper analysis, RAG-based question answering, academic paper search, recommendations, and knowledge-graph-based research exploration in a single platform. It goes beyond simple summarization by helping users understand papers, connect related research, and identify potential research directions.

2.Problem statement, objectives, and scope

Problem Statement

The increasing number of research papers makes it difficult for students and researchers to efficiently search, understand, and analyze academic literature. Existing platforms often provide only search or basic summarization, requiring multiple tools for research activities. Therefore, there is a need for an AI-powered platform that supports research-paper analysis, understanding, and management in one environment.

The main objectives of ScholarMind-AI are:

- To develop an AI-powered platform for assisting students and researchers throughout the research process.
- To provide detailed AI-based research paper summarization and analysis.
- To extract important information such as research objectives, methodology, key findings, limitations, and future scope.

- To enable users to ask questions about uploaded research papers using RAG-based responses.

2.Problem statement, objectives, and scope

- To provide academic paper search and personalized research recommendations.
- To organize papers through projects, bookmarks, history, and a personal research library.
- To explore relationships between papers, authors, topics, and methodologies using knowledge graphs.
- To assist researchers in identifying potential research gaps and emerging research trends.

Scope

The scope of ScholarMind AI is to provide a centralized AI-powered platform for academic research. It enables users to search, upload, analyze, understand, and organize research papers using AI technologies. The system supports detailed paper analysis, RAG-based question answering, academic search, project management, bookmarks, and personalized recommendations. It can also help users explore relationships between research papers, authors, and topics, providing a more efficient and organized approach to academic research.

3. Literature review

S.no	Paper & Authors	Year	Area	Key Contribution	Limitation / Gap
1.	SPECTER: Document-level Representation Learning using Citation-informed Transformers — Arman Cohan et al	2020	Scientific Document Representation	Generates scientific paper embeddings using text and citation information for semantic comparison and recommendation.	Focuses on document representation rather than complete paper analysis and knowledge discovery.
2.	Scientific Paper Recommendation: A survey — Xiaomei Bai et al	2024	Knowledge Graph & Recommendation	Combines paper content, citations, author relationships, and user behavior for personalized recommendations..	Focuses mainly on recommendation accuracy rather than detailed paper analysis.
3.	Multi-Feature-Enhanced Academic Paper Recommendation Model with Knowledge Graph — Le Wang et al	2024	Knowledge Graph & Recommendation	Combines paper content, citations, author relationships, and user behavior for personalized recommendations.	Focuses mainly on recommendation accuracy rather than detailed paper analysis..
4.	Recommending Papers Based on Academic knowledge Graph and Biased Random Walk-	2024	Knowledge Graph & Recommendation	Uses academic knowledge graphs and citation information to identify papers related to researchers' interests.	Does not provide detailed paper summarization or comprehensive knowledge extraction.
5.	Academic Literature Recommendation in Large-scale Citation Networks Enhanced by Large Language Models — Kun Liu et al.	2024	LLM & Academic Recommendation	Combines citation networks, semantic embeddings, and user preferences for personalized literature recommendation.	Focuses on recommendation rather than integrated research-paper analysis and knowledge discovery.

4.Research gaps and proposed solution

Research Gap

These papers collectively show that significant research has been conducted in semantic scientific-document representation, academic recommendation, citation networks, knowledge graphs, and personalization. However, the capabilities are generally addressed as separate components.Existing systems focuses on scientific document representations, while recommendation studies primarily focus on ranking relevant papers using content, citations, graphs, or user preferences.

Proposed Solution

The proposed solution, ScholarMind AI, integrates research-paper analysis, semantic understanding, knowledge discovery, and academic recommendation. It extracts key information, identifies relationships among papers and concepts, generates meaningful insights, and recommends relevant literature based on researchers' interests, reducing literature-review effort and improving academic research efficiency.

5.Software requirements (Software and Hardware requirements)

Software Requirements

Frontend: React.js, Vite, JavaScript, Tailwind CSS

Backend: Python, FastAPI, Uvicorn

RAG: LangChain, Embeddings, Vector Search, pgvector

Database: PostgreSQL

AI: Google Gemini API

Tools: VS Code, Git, GitHub

Hardware Requirements

Processor: Intel Core i5 or equivalent

RAM: 8 GB minimum

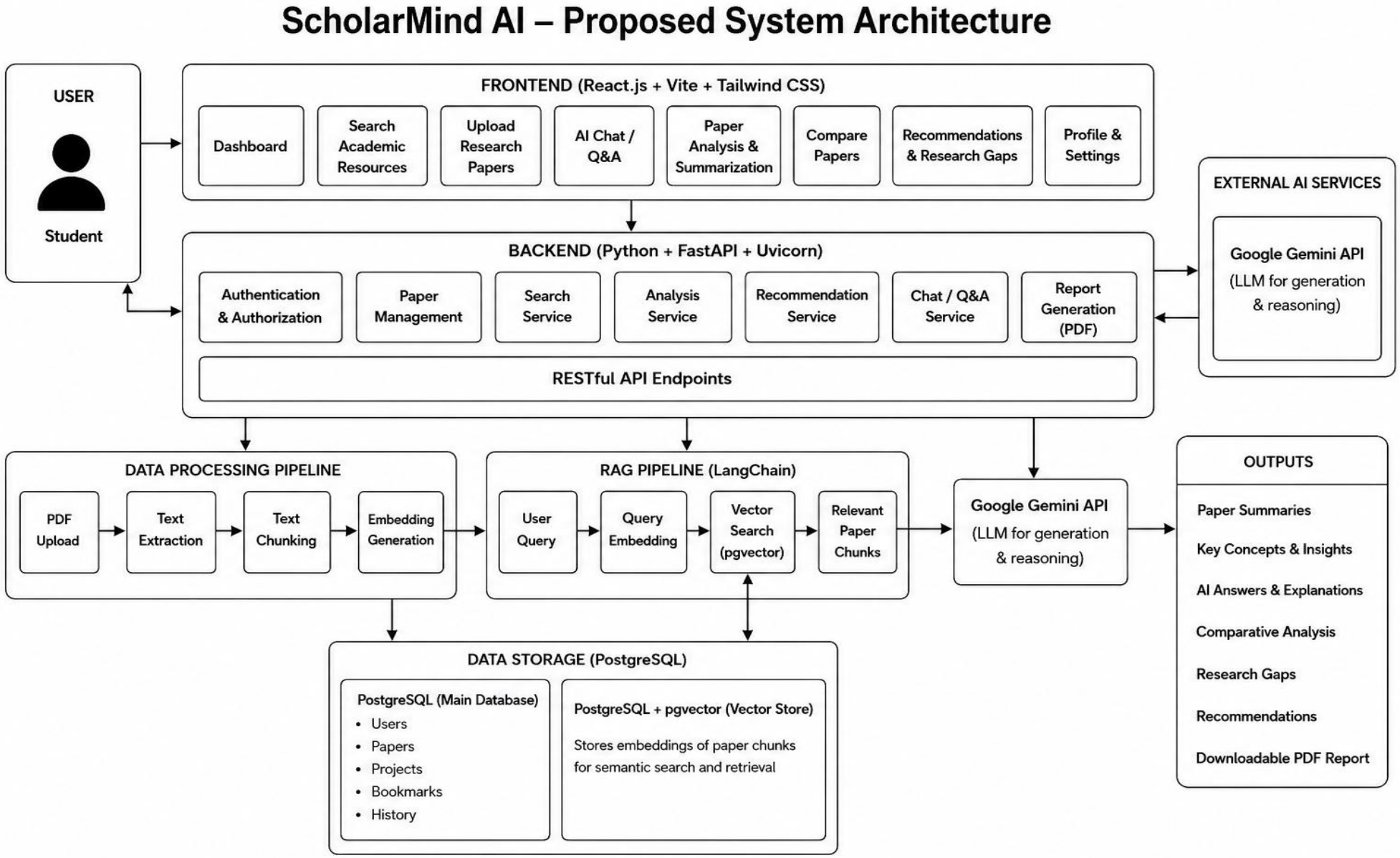
Storage: 256 GB SSD

Display: 1366 × 768 or higher

Internet: Stable Internet Connection

GPU: Not required

7.Design and overall system architecture.



References

- **Research Papers**
- Cohan, A., Feldman, S., Beltagy, I., Downey, D., & Weld, D. S. (2020). *SPECTER: Document-level Representation Learning using Citation-informed Transformers*. **ACL 2020**.
- Bai, X., Wang, M., Lee, I., Yang, Z., Kong, X., & Xia, F. (2020). *Scientific Paper Recommendation: A Survey*. **IEEE Access**.
- Hui, L., Cui, Z., & Wang, Y. (2025). *Recommending Papers Based on Academic Knowledge Graph and Biased Random Walk*. **Data Analysis and Knowledge Discovery**.
- **Chakraborty, T., Krishna, A., Singh, M., Ganguly, N., Goyal, P., & Mukherjee, A. (2016).** *FeRoSA: A Faceted Recommendation System for Scientific Articles*. **PAKDD 2016**.
- **Nair, A. M., Benny, O., & George, J. (2021).** *Content Based Scientific Article Recommendation System Using Deep Learning Technique*